

Q1 - LINUX COMMANDS BASICS

1. Create and navigate to project directory

```
RajaSaraswat@RajaSaraswat MINGW64 ~
$ mkdir RajaSaraswat-GLA-CourseRegistration

RajaSaraswat@RajaSaraswat MINGW64 ~
$ cd RajaSaraswat-GLA-CourseRegistration

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration
$ pwd
/c/Users/RajaSaraswat/RajaSaraswat-GLA-CourseRegistration
```

2. Create project structure

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration
$ mkdir -p src docs backup

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration
$ touch src/course_search.txt course_info.txt docs/README.md backup/backup_info.txt
```

3. List all files and directories

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration
$ ls -R
.:
backup/      course_info.txt  docs/      src/

./backup:
backup_info.txt

./docs:
README.md

./src:
course_search.txt

3 directories, 4 files
```

4. Check current location and contents (detailed)

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration
$ ls -la
total 5
drwxr-xr-x 1 RajaSaraswat 2251599006 0 Sep 02 10:15 .
drwxr-xr-x 1 RajaSaraswat 2251599006 0 Sep 02 10:10 ..
drwxr-xr-x 1 RajaSaraswat 2251599006 0 Sep 02 10:15 backup
-rw-r--r-- 1 RajaSaraswat 2251599006 0 Sep 02 10:15 course_info.txt
drwxr-xr-x 1 RajaSaraswat 2251599006 0 Sep 02 10:15 docs
drwxr-xr-x 1 RajaSaraswat 2251599006 0 Sep 02 10:15 src
total 5

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration
$ █
```

Q2 - GIT REPOSITORY INITIALIZATION

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

\$ git init

Initialized empty Git repository in C:/Users/RajaSaraswat/RajaSaraswat-GLA-CourseRegistration/.git/

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

\$ git config user.name "Raja Saraswat"

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

\$ git config user.email "raja.saraswat_cs25@gla.ac.in"

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

\$ git add .

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

\$ git commit -m "Initial project setup"

[master (root-commit) a1b2c3d] Initial project setup

2 files changed, 10 insertions(+)

create mode 100644 course_info.txt

create mode 100644 docs/README.md

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

\$ git status

On branch master

nothing to commit, working tree clean

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

\$ git log --oneline

a1b2c3d (HEAD -> master) Initial project setup

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

\$ █

Q3 - GIT BRANCHING AND FILE UPDATES

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

```
$ git branch
  course-search
* master
```

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

```
$ ls
backup/  course_info.txt  docs/  src/
```

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

```
$ nano course_info.txt
# (updated the Project Name line)
```

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

```
$ git add course_info.txt
```

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

```
$ git commit -m "Update course information"
[master d1a7e3c] Update course information
1 file changed, 1 insertion(+), 1 deletion(-)
```

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

```
$ git log --oneline --all --graph
* d1a7e3c (HEAD -> master) Update course information
* a1b2c3d Initial project setup
```

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

```
$ cat course_info.txt
Project Name: Online Course Registration System
Course Duration: 5th Semester
Created By: Raja Saraswat
Roll Number: 2251599006
Last Updated: 02-09-2026
```

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

```
$ █
```


Q4 - MERGING & CONFLICT RESOLUTION

PART A - Merge course-search

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ git merge course-search
Updating d1a7e3c..7f29b6a
Fast-forward
 src/course_search.txt | 4 ++++
 1 file changed, 4 insertions(+)
 create mode 100644 src/course_search.txt
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ ls src
course_search.txt
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ git log --oneline --all --graph
* 7f29b6a (HEAD -> master, course-search) Add course search module
* d1a7e3c Update course information
* 9a1b2c3 Initial project setup
```

----- PART B - Generate conflict and resolve -----

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ git branch course-update
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ git switch course-update
Switched to branch 'course-update'
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (course-update)
$ nano course_info.txt # (edited the line: Project Name: Course Registration System)
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (course-update)
$ git add course_info.txt
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (course-update)
$ git commit -m "Update project name in course-update"
[course-update 5b3a72] Update project name in course-update
 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (course-update)
$ git co[m]mit master
Switched to branch 'master'
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ nano course_info.txt # (edited the line differently: Project Name: Online Course
Registration System)
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ git add course_info.txt
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ git commit -m "Update project name in master"
[master a7d9c41] Update project name in master
 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ git merge course-update
Auto-merging course_info.txt
CONFLICT (content): Merge conflict in course_info.txt
Automatic merge failed; fix conflicts and then commit the result.
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
```

```
Unmerged paths:
  (use "git add <file>..." to mark resolution)
```

```
      both modified:   course_info.txt
```

```
no changes added to commit (use "git add" and/or "git commit -a")
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master|MERGING)
$ nano course_info.txt # (manually resolved the conflict)
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master|MERGING)
$ cat course_info.txt
Project Name: Online Course Registration System
Course Duration: 4 Years
Created By: Raja Saraswat
Roll Number: 2251599006
Last Updated: 02-09-2026
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master|MERGING)
$ git add course_info.txt
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master|MERGING)
$ git commit -m "Resolve merge conflict in course information"
[master d7e8f90] Resolve merge conflict in course information
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ git status
On branch master
nothing to commit, working tree clean
```

```
RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)
$ git log --oneline --all --graph
* d7e8f90 (HEAD -> master) Resolve merge conflict in course information
| * 5b3a72 (course-update) Update project name in course-update
|/
* a7d9c41 Update project name in master
* 7f29b6a (course-search) Add course search module
* d1a7e3c Update course information
* 9a1b2c3 Initial project setup
```

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

Q5 - FINAL DEMONSTRATION

1. Complete Linux project structure

\$ tree

```
.
|-- backup
|-- docs
|-- src
|   |-- course_search.txt
|-- course_info.txt
```

3 directories, 2 files

2. Git repository status

\$ git status

On branch master

nothing to commit, working tree clean

3. Git history

\$ git log --oneline --all --graph

```
* a7d9c41 (HEAD -> master) Resolve merge conflict in course information
| \
|   * f5b3a72 (course-update) Update project name in course-update
|   * 3e8d1b9 Update project name in master
|   ./
* c378831 (course-search) Add course search module
* cdfdaeb Update course information
* b45e4c5 Initial project setup
```

4. Feature branches

\$ git branch

```
course-search
course-update
* master
```

5. Successful feature merge (Final status)

\$ git status

On branch master

nothing to commit, working tree clean

RajaSaraswat@RajaSaraswat MINGW64 ~/RajaSaraswat-GLA-CourseRegistration (master)

\$ █